

EEG report labeling with MedGemma — five findings, and how they compare to Harvard

This pulls together everything we’ve labeled so far: two slowing findings (focal, generalized) and three background patterns (discontinuous, burst-suppression, suppressed). Each is a separate field with a three-way call — the report says it’s there (present), says it isn’t (explicitly_absent), or doesn’t bring it up (not_mentioned). Everything ran on Fraser Health (45,545 reports) and three Harvard hospitals — MGH, BWH, BCH (~101,600; BIDMC has no text) — with MedGemma-27B (Q4_K_S), temperature 0, grammar-constrained so every answer is one clean status.

Three of the five findings have a matching column in the Harvard database and can be compared head-to-head: focal slowing (foc slowing), generalized slowing (gen slowing), and burst-suppression (bs). Discontinuous and suppressed background have no Harvard variable, so for those we only have our own labels.

The comparison, in one table

Harvard’s value is present-or-blank and carries a *provenance* (was the finding taken from the report text, from the recording’s annotations, or expert-verified). We compare three ways: against everything they marked, against only what they took from the report text (the fair like-for-like, since we read only the text), and — the closest thing to a right answer — our recall on the findings an expert had verified. How their values are coded is spelled out in the companion note reports/harvard_llm_labeling.md.

field — hospital	ours present	Harvard present	agreement	κ	κ text-only	our recall on verified
focal slowing — MGH	34.5%	14.1%	77%	0.40	0.37	87% (1480)
focal slowing — BWH	50.6%	23.3%	70%	0.41	0.39	92% (413)
focal slowing — BCH	17.7%	7.4%	87%	0.43	0.42	—
generalized slowing — MGH	60.6%	77.2%	74%	0.40	0.47	71% (3743)
generalized slowing — BWH	50.5%	72.2%	65%	0.30	0.37	78% (1243)
generalized slowing — BCH	30.6%	27.4%	84%	0.60	0.66	—
burst-suppression — MGH	3.5%	11.2%	92%	0.44	0.60	65% (231)
burst-suppression — BWH	3.0%	13.8%	89%	0.32	0.50	61% (184)
burst-suppression — BCH	1.3%	8.5%	93%	0.25	0.29	—

Reading it: *agreement* is how often both sides give the same present/not call; κ is that agreement after removing what chance alone would give (0 = chance, 1 = perfect). κ around 0.3–0.6 is moderate. κ *text-only* compares against just the labels Harvard took from the report text. *Recall on verified* is how

often we catch the findings an expert confirmed (positives only; BCH has none).

The patterns are consistent and mostly explainable:

- **Focal slowing — we call it more than they do** (about 2×), yet on the expert-verified cases we catch ~87–92%. So the extra is largely real findings the model reads out of the text.
- **Generalized slowing — they call it more than we do**, and we miss ~1 in 4 verified cases, so our under-calling is partly genuine. Comparing text-to-text lifts κ (their annotation-only labels aren't in the narrative).
- **Burst-suppression — their bs is marked 3-6× more than ours**, at MGH/BWH almost entirely through their annotation channel: text-to-text κ jumps to 0.50–0.60 there. Reading only the report, we agree well with what the *report* says; we just don't see the annotation-only flags. BCH is the exception — there their bs is itself mostly from the text, so the gap is a genuine under-call on our side, not an annotation artifact (text-only κ stays low, 0.29).

Charts

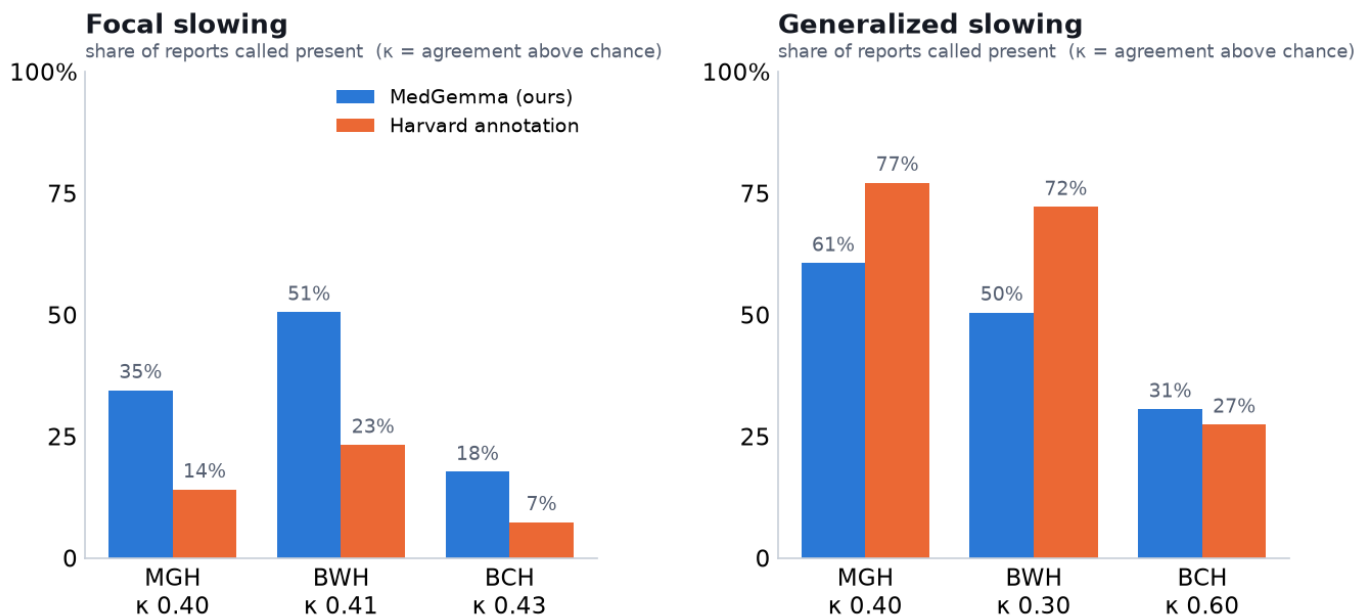


Figure 1: Focal / generalized slowing — present rate, ours vs Harvard

Fraser Health

Fraser Health has no labels of its own for any of these, so here we just have what the model found:

field	present	explicitly_absent	not_mentioned
focal slowing	25.6%	24.5%	50.0%
generalized slowing	23.0%	40.2%	36.8%
discontinuous	0.1%	1.8%	98.1%
burst-suppression	1.1%	0.1%	98.8%
suppressed	1.0%	1.7%	97.4%

The background patterns are rare, as expected for ICU / anesthesia findings. Discontinuous stands out: Fraser Health reports almost never comment on continuity (0.1% present, 98% not mentioned), whereas the Harvard reports do so often — at BCH, 63% of reports explicitly call the background continuous. That's a difference in how the two cohorts write reports, not a disagreement about any one recording.

Burst-suppression: ours vs Harvard

share called present (κ = agreement above chance)

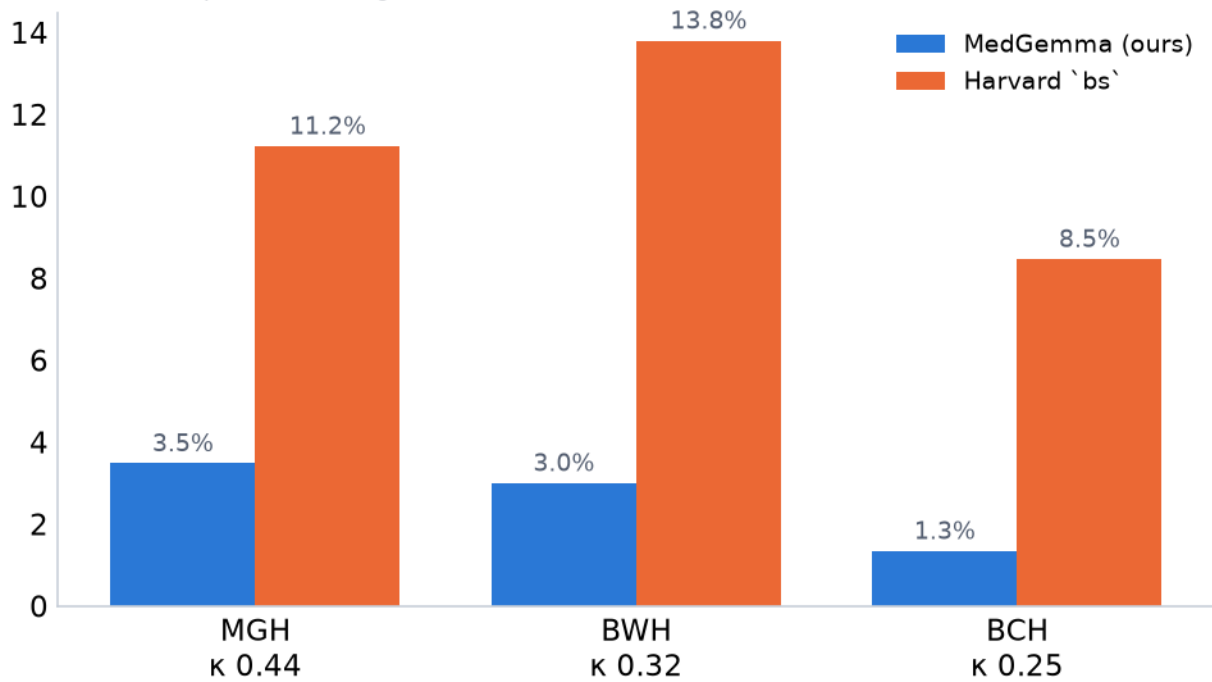


Figure 2: Burst-suppression — ours vs Harvard

Do we catch what an expert confirmed?

our recall on Harvard's expert-verified present reports

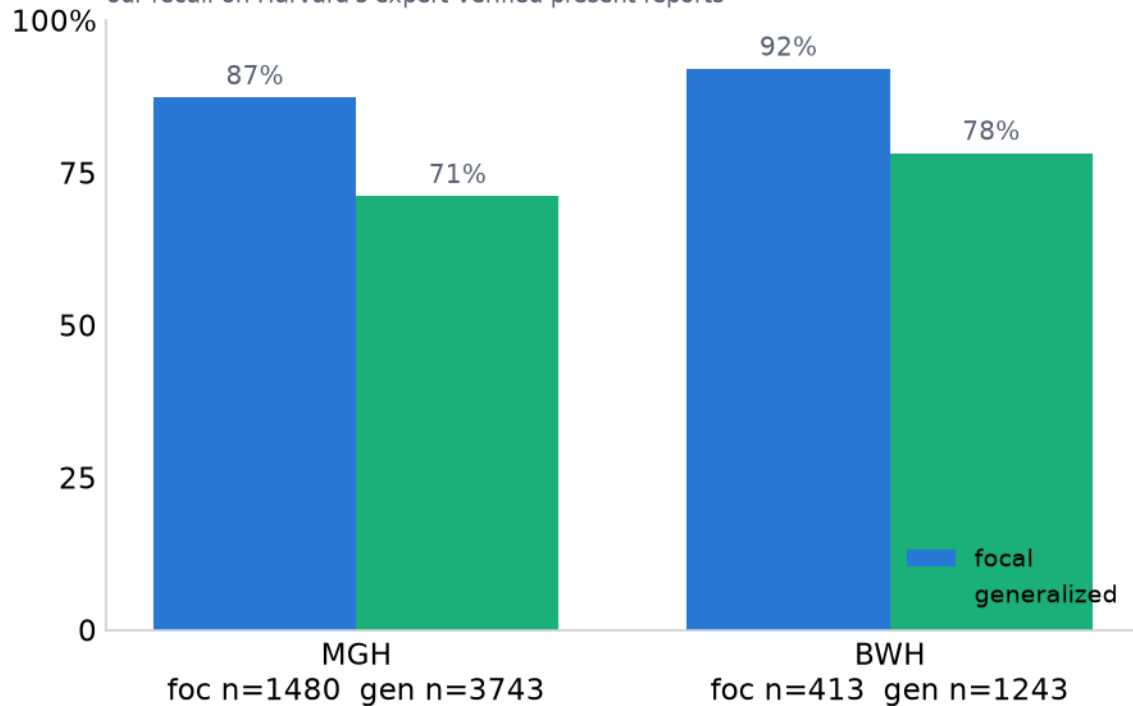


Figure 3: Our recall on expert-verified slowing

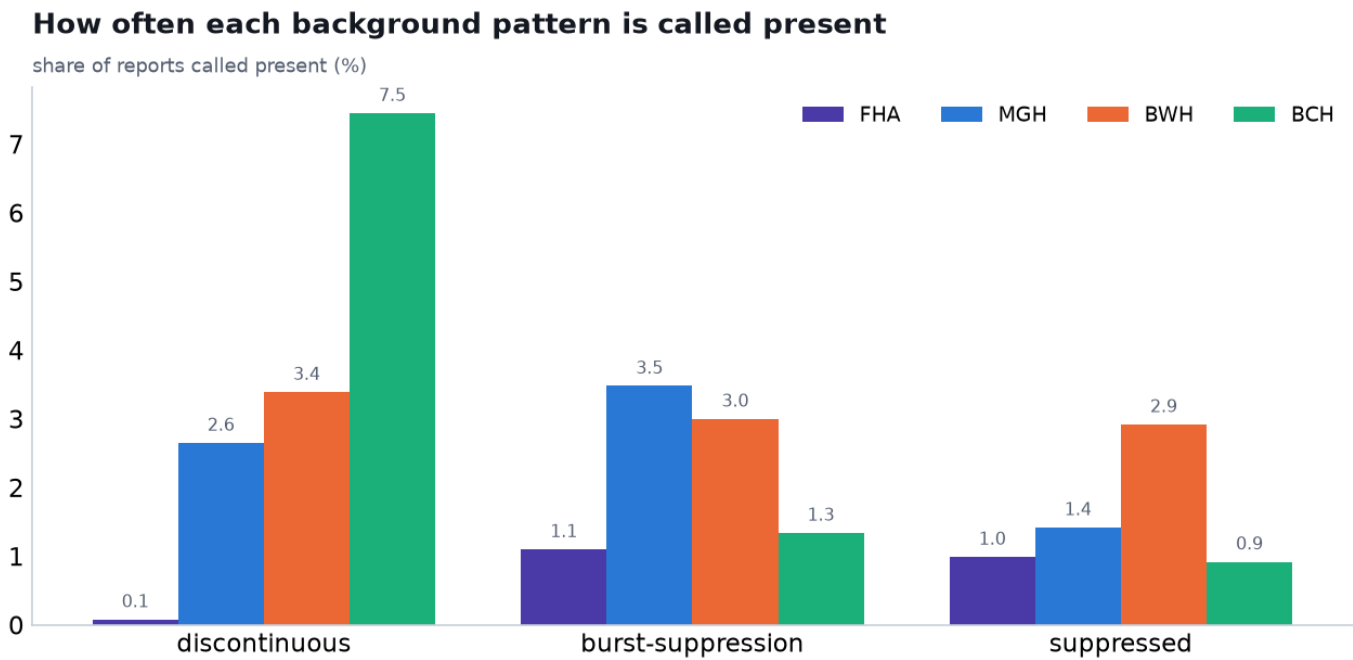


Figure 4: Prevalence of the three background patterns across datasets

Discontinuous and suppressed background

Neither has a Harvard column to check against (the nearest, low voltage, is a different thing), so these stay labels-only. Present rates are low throughout — discontinuous 2.6–7.5% across the Harvard hospitals, suppressed 0.9–2.9%.

Caveats

- About 1.3–1.4% of Harvard reports are too long for the context window and returned no label; they're excluded from the counts.
- Harvard's column is present-or-blank; a blank mixes "the report says absent" with "not mentioned" and "not annotated". We keep those apart on our side but collapse to present-vs-not to compare.
- The verified check is positives-only (no verified "absent") and has no BCH data, so it measures recall, not precision. A small hand-labeled set with negatives would let us score both models fully — including Fraser Health, which has no ground truth today.

Reproduce

```
# labeling (on the cluster)
bash slurm/submit_slowing.sh ; bash slurm/submit_slowing_harvard.sh
bash slurm/submit_background.sh ; bash slurm/submit_background_harvard.sh
# comparisons, distributions, charts, table
python -m analysis.slowing_vs_harvard
python -m analysis.background_vs_harvard
python -m analysis.background_dist
python -m analysis.slowing_chart ; python -m analysis.background_chart
python -m analysis.master_table
```

Companion notes: reports/harvard_llm_labeling.md (how Harvard's fields and values are coded), reports/slowing_vs_harvard.md and reports/background_patterns.md (the per-topic write-ups).